DECOMPOSITIONS · QUADRATIC FORMS

Definiteness, read from the signs

The quadratic form $\mathbf{x}^{\mathsf{T}} A\mathbf{x}$ is classified entirely by the signs of the eigenvalues — not their sizes. Five classes, and the surface each one describes follows immediately.

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CLASSES

TYPE	CONDITION	EIGENVALUE SIGNS	SIGN OF THE FORM	LEVEL SURFACE
ONE SIGN THROUGHOUT				
 **Positive definite**	all $\lambda_i > 0$	all strictly positive	> 0 for every $\mathbf{x} \neq \mathbf{0}$	ellipsoid
 **Negative definite**	all $\lambda_i < 0$	all strictly negative	< 0 for every $\mathbf{x} \neq \mathbf{0}$	inverted ellipsoid
ONE SIGN, WITH ZEROS				
 **Positive semi-definite**	all $\lambda_i \geq 0$, at least one $= 0$	non-negative, some zero	≥ 0; vanishes on the null space	degenerate ellipsoid — cylinder
 **Negative semi-definite**	all $\lambda_i \leq 0$, at least one $= 0$	non-positive, some zero	≤ 0; vanishes on the null space	inverted cylinder
BOTH SIGNS PRESENT				
 **Indefinite**	some $\lambda_i > 0$ and some < 0	mixed	takes both signs	saddle — hyperboloid

In the eigenbasis the form becomes $\sum \lambda_i y_i^2$, and the sign question answers itself: a sum of squares with all-positive weights is positive, with mixed weights it takes both signs. Everything below is that substitution read off. In practice Cholesky tests definiteness faster than computing the eigenvalues does.